Phase 1 PAT

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Summary:

The program will be a divers logging system, where they can log into the program so that they can create, edit, view, and delete records of SCUBA dives with all relevant details (Dive length, weight used, diver number, depth etc.). They will also be able to search and sort dives based on certain criteria for ease of viewing and keeping records. This digital system will make logging and tracking of dives more efficient and less likely to be lost or damaged because it is stored digitally. The program will use a database to store all the information. The usernames and passwords will also be stored in a text file.

Program functions:

The program will:

- Allow users to log into the account using their username and password, to edit their records
- Allow them to create a new dive log
- Allow them to edit previously logged dives
- Allow them to view their previously logged dives
- Allow them to delete logged dives from their record
- Allow them to sort their dive log based on different criteria
- Allow them to search for specific dives
- Allow them to view different pages to perform different operation

The program will not:

- Convert dive logs from any form of digital diving device (such as a dive computer)
- Calculate decompression time
- Analise dive information

Specifications of user interface:

The user interface will match the function of the program in that the color scheme and other factors will be ocean themed and oriented. The user will be able to navigate between panels using tabs for ease of use. There will be an initial log in panel, thereafter there will be a separate panel to create a new dive log as there is a lot of information to enter and a separate panel to view previously logged dives. It will be easy to navigate but still perform many complex functions. The program will be designed to make a diver's life easy and efficient, providing them with a long-term storage option for their dive log using persistent storage.

Specifications of data storage:

The program will use a database table to store the information and a text file to store usernames and passwords.

tblLog:

- DiveNumber [integer] - Primary key uniquely identifying the dive
- DiveDate [Date] - the date on which the dive took place
- TimeIn [Time] - The time that the dive commenced
- TimeOut [Time] - The time that the dive ended
- AirIn [Integer] - The amount of air in the tank at the start of the dive
- AirOut [Integer] - The amount of air in the tank at the end on the dive
- Weight [Integer] - The weight used by the diver in kgs
- Depth [Double] - The maximum depth reached on the dive
- Temp [Double] - The water temperature at the time of the dive
- Visibility [Integer] - The distance visible under the water
- Weather [String] - The weather conditions at the time of the dive
- Buddy [String] - The name of the person partnered with the diver on the dive
- BuddyCertNumber [Integer] - The dive number of the dive partner on the dive
- Comments [String] - Any additional information
- SafetyStop [Boolean]– whether the diver made the recommended safety stop on the dive

Text File:

Each line will contain a username and password in the format of Username#Password

Hardware and software requirements:

Hardware	Software
• 3.52 GHz ryzen_3_3200u CPU • 1 X 4 GB DDR4 RAM 3.5 GHz • AMD Radeon Vega 3 integrated Graphics • 128 GB Flash Memory SSD • Acer Aspire 5 with above mentioned hardware requirements	• Apache NetBeans IDE 11.2 – used to develop the program • Java JDK 13.01 – language used to code the program • Access 2016 – Used to create and manage the database • Word 2016 – Used to plan the program • Notepad – Used to create and store the text file which stores the username and passwords • Windows 10 pro 21H1 – Operating system used to run the device on which the program will be made